

ACUTE Lab Website — Content Management Guide

This guide explains how to use the ACUTE Content Manager (CMS) to edit the lab website without touching code.

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1. Getting Started

Prerequisites

You need the following installed on your computer:

- **Node.js** (version 20 or higher) — [download here](#) (Windows: use the `.msi` installer, keep all defaults)
- **Git** — [download here](#) (Windows: install "Git for Windows", keep all defaults — this includes Git Bash)
- **SSH key** — needed to publish changes to GitHub. See [Section 2: Setting Up SSH](#)
- **Ruby + Jekyll** — only needed for the Preview feature (optional, see [Section 13](#))

Which terminal to use

Operating System	Terminal
Windows	Open Git Bash (installed with Git for Windows). Right-click on your desktop and select "Git Bash Here", or search for "Git Bash" in the Start menu.
macOS	Open Terminal (Applications > Utilities > Terminal)

All commands in this guide work in both Git Bash (Windows) and Terminal (macOS).

First-Time Setup

1. **Clone the repository** (only the first time):

```
cd ~/Documents
git clone git@github.com:Acute-hi-is/Acute-hi-is.github.io.git acute_website_v2
```

2. **Install CMS dependencies:**

```
cd acute_website_v2/cms
npm install
```

This only needs to be done once (or after updates to the CMS).

Starting the CMS

Every time you want to edit content:

```
cd ~/Documents/acute_website_v2
git pull origin main      # get the latest changes
cd cms
npm run dev               # start the CMS
```

Open your browser and go to <http://localhost:3000>

Stopping the CMS

Press **Ctrl+C** in the terminal.

2. Setting Up SSH for GitHub

To publish changes from the CMS, your computer needs an SSH key linked to your GitHub account. You only need to do this **once per computer**.

If you can already push to GitHub (test with `ssh -T git@github.com`), skip this section.

Step 1: Open your terminal

- **Windows:** Open **Git Bash** (search for it in the Start menu)
- **macOS:** Open **Terminal** (Applications > Utilities > Terminal)

Step 2: Check if you already have an SSH key

Type:

```
ls ~/.ssh/id_ed25519.pub
```

- If you see a file path, **you already have a key** — skip to Step 4
- If you see "No such file or directory", continue to Step 3

Step 3: Generate a new SSH key

Type (replace with your email):

```
ssh-keygen -t ed25519 -C "your-name@hi.is"
```

You will be asked three questions. **Press Enter for all three** to accept the defaults:

```
Enter file in which to save the key (/c/Users/YourName/.ssh/id_ed25519): [press Enter]
Enter passphrase (empty for no passphrase): [press Enter]
Enter same passphrase again: [press Enter]
```

You should see output like:

```
Your identification has been saved in /c/Users/YourName/.ssh/id_ed25519
Your public key has been saved in /c/Users/YourName/.ssh/id_ed25519.pub
```

Step 4: Copy your public key

Type:

```
cat ~/.ssh/id_ed25519.pub
```

This prints a long line starting with `ssh-ed25519 AAAA...` and ending with your email.

Select the entire line and copy it:

- **Windows (Git Bash):** select with mouse, right-click > Copy (or Ctrl+Insert)
- **macOS:** select with mouse, Cmd+C

Step 5: Add the key to GitHub

1. Open your browser and go to: <https://github.com/settings/keys>
 - If you're not logged in, log in first
2. Click the green **New SSH key** button
3. Fill in:
 - **Title:** a name for this computer (e.g. "Lab laptop" or "Office PC")
 - **Key type:** keep as "Authentication Key"
 - **Key:** paste the key you copied in Step 4
4. Click **Add SSH key**
5. GitHub may ask you to confirm your password

Step 6: Test the connection

Back in your terminal, type:

```
ssh -T git@github.com
```

If this is your first time connecting, you'll see:

```
The authenticity of host 'github.com' can't be established.
Are you sure you want to continue connecting (yes/no)?
```

Type **yes** and press Enter.

You should then see:

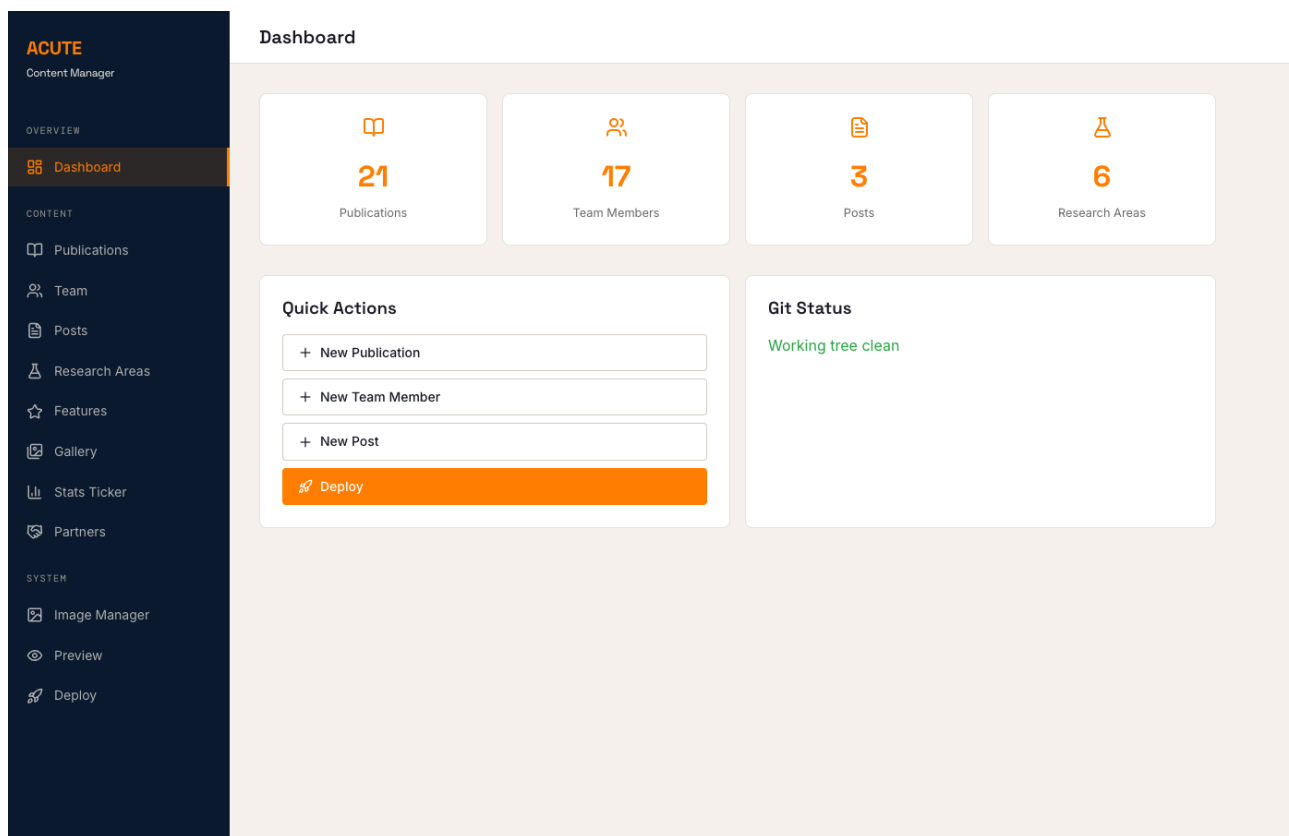
```
Hi YourUsername! You've successfully authenticated, but GitHub does not provide shell access.
```

If you see this message, **you're done**. You can now push changes from the CMS.

If it doesn't work, see [Section 15: Troubleshooting](#).

3. The Dashboard

When you open the CMS, you land on the **Dashboard**.



The dashboard shows:

- **Content counts** — how many publications, team members, posts, and research areas exist
- **Quick Actions** — shortcuts to create new content or deploy changes
- **Git Status** — whether you have unsaved (uncommitted) changes

4. Managing Publications

Click **Publications** in the sidebar.

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+ Add

AllHapticsAcousticsPerception

21 publications

TITLE	YEAR	TOPIC	DOI	
Discriminating Vibrotactile Signals: The Relative Roles of Amplitude and Frequency I. Makarov, Á. Kristjánsson, R. Unnthorsson	2026	HAPTICS		
Perceptual Design and Evaluation of a Forearm-Based Vibrotactile Interface for Transfemoral Prosthetic Feedback M. Karimi, S. Brynjólfsson, K. Briem, Á. Kristjánsson, R. Unnthorsson	2026	HAPTICS		
Stable Model Reduction for Time-Domain Room Acoustics: A Structure-Preserving Formulation for Complex Boundaries S. Bonthu, H. Sampedro Llopis, S. Thrastarson, F. Pind, R. Unnthorsson	2026	ACOUSTICS		
Sensory Substitution Through Vibrotactile Stimulation Á. Kristjánsson, I. Makarov, N. Yeganeh, R. Unnthorsson	2026	HAPTICS		
Vibrotactile Pattern Recognition: Influence of Interstimulus Intervals N. Yeganeh, I. Makarov, Á. Kristjánsson, R. Unnthorsson	2025	HAPTICS		
Design and Manufacture of Synthetic Pinnæ for Studying Head-Related Transfer Functions (HRTFs) E. M. Sumner, M. Riedel, R. Unnthorsson	2025	ACOUSTICS		
Haptic Feedback Systems for Lower-Limb Prosthetic Applications: A Review of System Design, User Experience, and Clinical Insights M. Karimi, N. Yeganeh, I. Makarov, Á. Ö. Sverrisson, K. F. Gunnarsson, K. Briem, S. Brynjólfsson, Á. Kristjánsson, R. Unnthorsson	2025	HAPTICS		
Cross-Modal Cues Improve the Detection of Synchronized Targets During Human Foraging I. Makarov, R. Unnthorsson, Á. Kristjánsson, I. M. Thornton	2024	PERCEPTION		

What you see

- A list of all publications sorted by year (newest first)
- **Filter buttons** at the top: All, Haptics, Acoustics, Perception
- Each row shows the title, authors, year, topic badge, and a DOI link

Adding a new publication

1. Click the orange **+ Add** button (top right)
2. Fill in the form:

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New Publication

TITLE

AUTHORS

VENUE

YEAR

2026

DOI

TOPIC

Haptics

SUMMARY

IMAGE URL (OPTIONAL)

PDF PATH (OPTIONAL)

Save

Cancel

Field	What to enter
Title	Full paper title
Authors	Author names, comma-separated (e.g. "I. Makarov, R. Unnthorsson")
Venue	Journal or conference name
Year	Publication year
DOI	The DOI identifier (e.g. "10.3390/s25010001")
Topic	Select: Haptics, Acoustics, or Perception
Summary	A 1-2 sentence description of the paper
Image URL	Optional — path to a thumbnail image
PDF Path	Optional — path to a locally hosted PDF

3. Click **Save**

Editing a publication

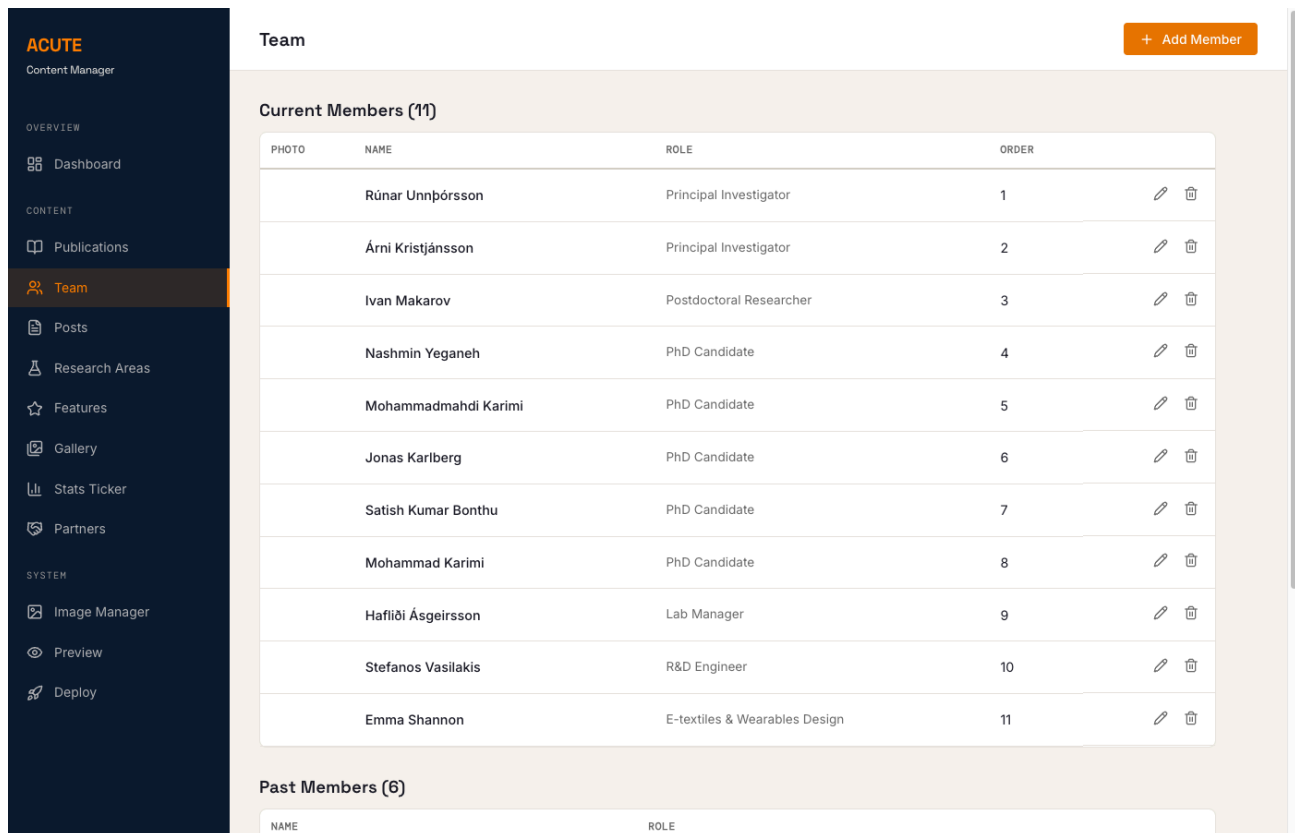
Click the pencil icon on any row. The form opens with the current values. Make your changes and click **Save**.

Deleting a publication

Click the trash icon on any row. A confirmation dialog appears — click **Delete** to confirm.

5. Managing Team Members

Click **Team** in the sidebar.



The screenshot shows the ACUTE Content Manager interface. The sidebar on the left has a dark blue background with the ACUTE logo and navigation links: Overview, Content, and System. The main area is titled 'Team' and shows a list of 11 current members and 6 past members. The 'Current Members (11)' section contains a table with columns for Photo, Name, Role, and Order. The 'Past Members (6)' section is partially visible below.

PHOTO	NAME	ROLE	ORDER
	Rúnar Unnþórsson	Principal Investigator	1
	Árni Kristjánsson	Principal Investigator	2
	Ivan Makarov	Postdoctoral Researcher	3
	Nashmin Yeganeh	PhD Candidate	4
	Mohammadmahdi Karimi	PhD Candidate	5
	Jonas Karlberg	PhD Candidate	6
	Satish Kumar Bonthu	PhD Candidate	7
	Mohammad Karimi	PhD Candidate	8
	Hafliði Ásgeirsson	Lab Manager	9
	Stefanos Vasilakis	R&D Engineer	10
	Emma Shannon	E-textiles & Wearables Design	11

Members are grouped into **Current Members** and **Past Members**.

Adding a new team member

1. Click **+ Add Member** (top right)
2. Fill in:
 - **Name** — full name
 - **Role** — e.g. "PhD Candidate", "Postdoctoral Researcher"
 - **Photo** — click Upload to add a photo (auto-compressed to 400x400)
 - **Email** — optional
 - **Profile URL** — optional link to university page
 - **Status** — "current" or "past"
 - **Order** — display order (lower numbers appear first)
 - **Bio** — write in the text area. This supports Markdown formatting
3. Click **Save**

Moving a member to "Past"

Edit the member, change **Status** from "current" to "past", and save.

Editing a bio

Click the pencil icon on any member. The bio field supports Markdown:

- ****bold text**** for bold

- `*italic text*` for italic
- `[link text](url)` for links
- Blank lines for new paragraphs

6. Managing Blog Posts

Click **Posts** in the sidebar.

The screenshot shows the ACUTE Content Manager interface. The sidebar on the left has a dark blue background with the ACUTE logo and 'Content Manager' text. It lists sections: OVERVIEW (Dashboard, Publications, Team), CONTENT (Posts, Research Areas, Features, Gallery, Stats Ticker, Partners), and SYSTEM (Image Manager, Preview, Deploy). The 'Posts' section is highlighted. The main area, titled 'Posts', has a light beige background and a '+ New Post' button. It contains a table with three columns: TITLE, DATE, and CATEGORY. The table lists three posts, each with a title, a brief description, a date, a category (LAB or PUBLICATION), and edit/delete icons.

TITLE	DATE	CATEGORY
ACUTE Lab launches new website Our redesigned website is now live — featuring an overview of our research areas, team profiles, and a comprehensive publication archive.	2026-04-06T00:00:00.000Z	LAB
New paper: Stable model reduction for time-domain room acoustics Satish Bonthu's paper on structure-preserving model-order reduction for room acoustics has been published in the International Journal for Numerical Methods in Engineering.	2026-03-15T00:00:00.000Z	PUBLICATION
Prosthetic feedback interface published in Biomimetics Mohammadmahdi Karimi's work on a forearm-based vibrotactile interface for transfemoral prosthetic feedback is now published.	2026-02-20T00:00:00.000Z	PUBLICATION

Adding a new post

1. Click **+ New Post** (top right)
2. Fill in:
 - **Title** — the post headline
 - **Date** — publication date
 - **Category** — e.g. "publication", "lab", "event"
 - **Excerpt** — a short summary (shown on the homepage)
 - **Content** — the full post body (Markdown supported)
3. Click **Save**

The post filename is auto-generated from the date and title.

7. Managing Research Areas

Click **Research Areas** in the sidebar.

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+ Add Area

#	TITLE	SUMMARY	PUBS	
01	Vibrotactile Perception	Mapping the frequency, amplitude, and spatial limits of touch on the forearm and wrist.	3	
02	Prosthetic Haptic Feedback	Vibrotactile sleeves that restore proprioception for transfemoral amputees.	2	
03	Sensory Substitution	Devices that translate sound and vision into touch for visually and hearing-impaired users.	2	
04	Spatial Audio & HRTFs	Synthetic pinna models and machine learning to personalise 3D sound.	3	
05	Room Acoustics Simulation	Structure-preserving model reduction achieving 100× speedup in wave simulation.	1	
06	Attention, Foraging & Haptic Illusions	Cross-modal foraging, visual synchrony effects, and haptic illusions.	2	

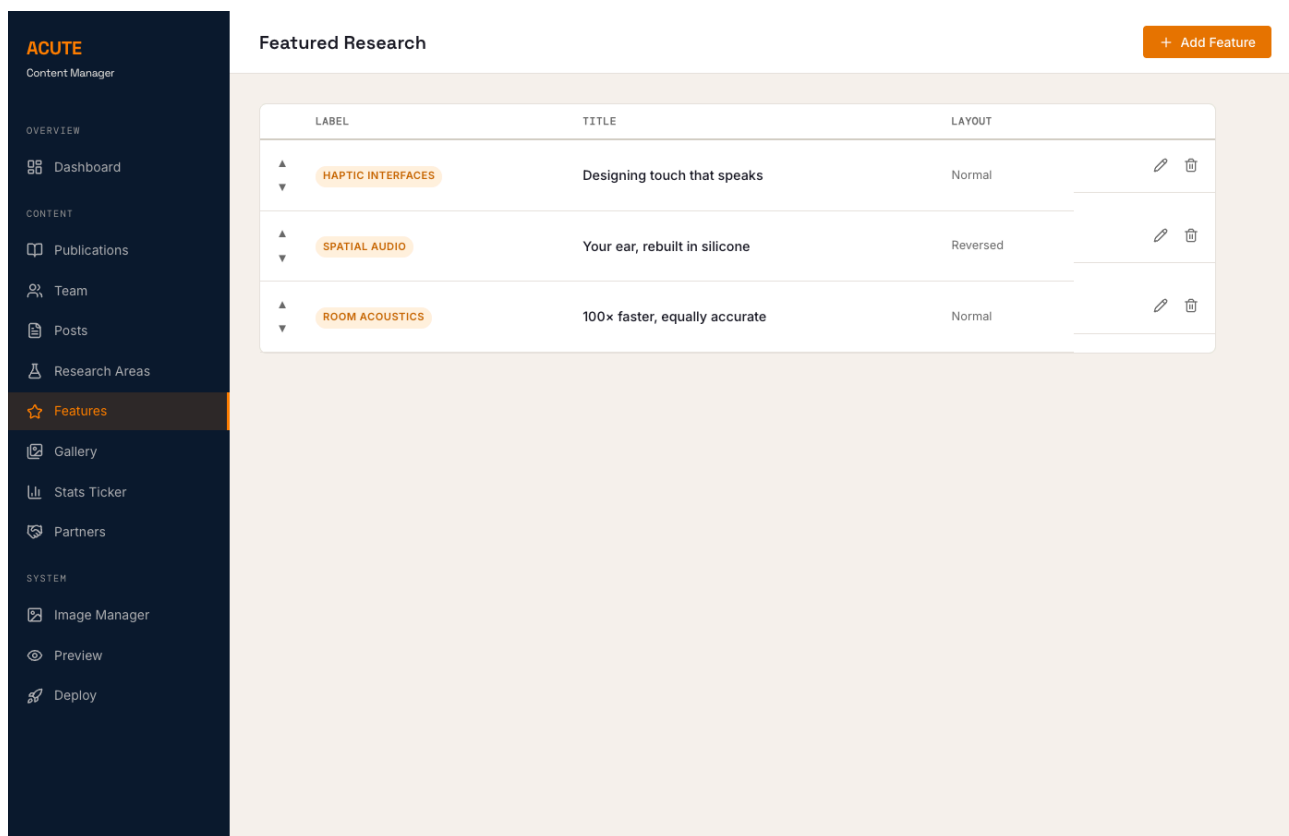
Each research area has a title, summary, description paragraphs, and linked publications.

Editing a research area

1. Click the pencil icon on any row
2. The form includes:
 - **Title** and **Summary**
 - **Description** — multiple paragraphs (use Add/Remove buttons to manage)
 - **Publications** — linked papers with text and DOI (use Add/Remove buttons)
 - **Images** — main image and highlight image
3. Click **Save**

8. Managing Featured Research

Click **Features** in the sidebar.



These are the highlighted research sections on the homepage.

Reordering

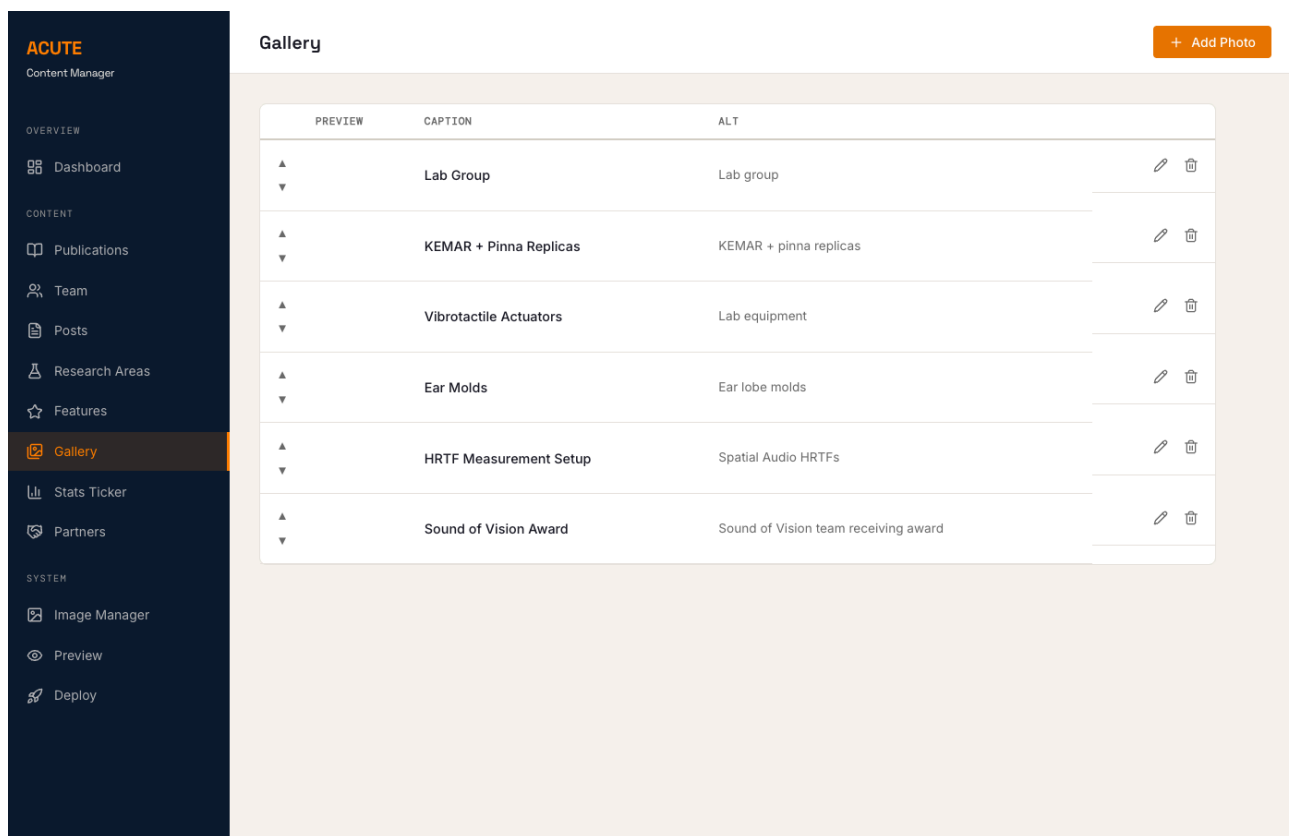
Use the up/down arrows on the left of each row to change the display order.

Editing a feature

Click the pencil icon. You can change the label, title, description text, DOI link, image, and layout direction (Normal or Reversed).

9. Managing the Gallery

Click **Gallery** in the sidebar.



The gallery controls the photo carousel on the homepage.

Adding a photo

1. Click **+ Add Photo** (top right)
2. Set the **image path** (type it or use Upload)
3. Add a **caption** and **alt text** (for accessibility)
4. Click **Save**

Reordering photos

Use the up/down arrows to change the carousel order.

10. Managing the Stats Ticker

Click **Stats Ticker** in the sidebar.

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+ Add Stat

NUMBER	LABEL	
▲ ▼ 93%	Sequential vibrotactile pattern recognition accuracy on forearm	
▲ ▼ 72.4%	Sequential vs 42.7% simultaneous accuracy for prosthetic joint feedback	
▲ ▼ ~93%	Ceiling accuracy at 240 ms signal duration for prosthetic feedback	
▲ ▼ 92-98%	Peak short-pattern accuracy at 300 ms interstimulus interval	
▲ ▼ 95%	Vibrotactile discrimination accuracy near wrist & elbow anchors	
▲ ▼ 34 dB	Individual variation range in wrist vibrotactile thresholds	
▲ ▼ 200 Hz	Peak wrist sensitivity frequency (inner/ glabrous skin)	
▲ ▼ 50 dB	Dynamic range available in the 50-300 Hz band at the inner wrist	
▲ ▼ 30	Participants in wrist threshold study across 25-1,000 Hz	
▲ ▼ 14	Braille-like patterns tested on a 2x3 actuator forearm array	

The stats ticker is the scrolling numbers bar on the homepage.

Adding a stat

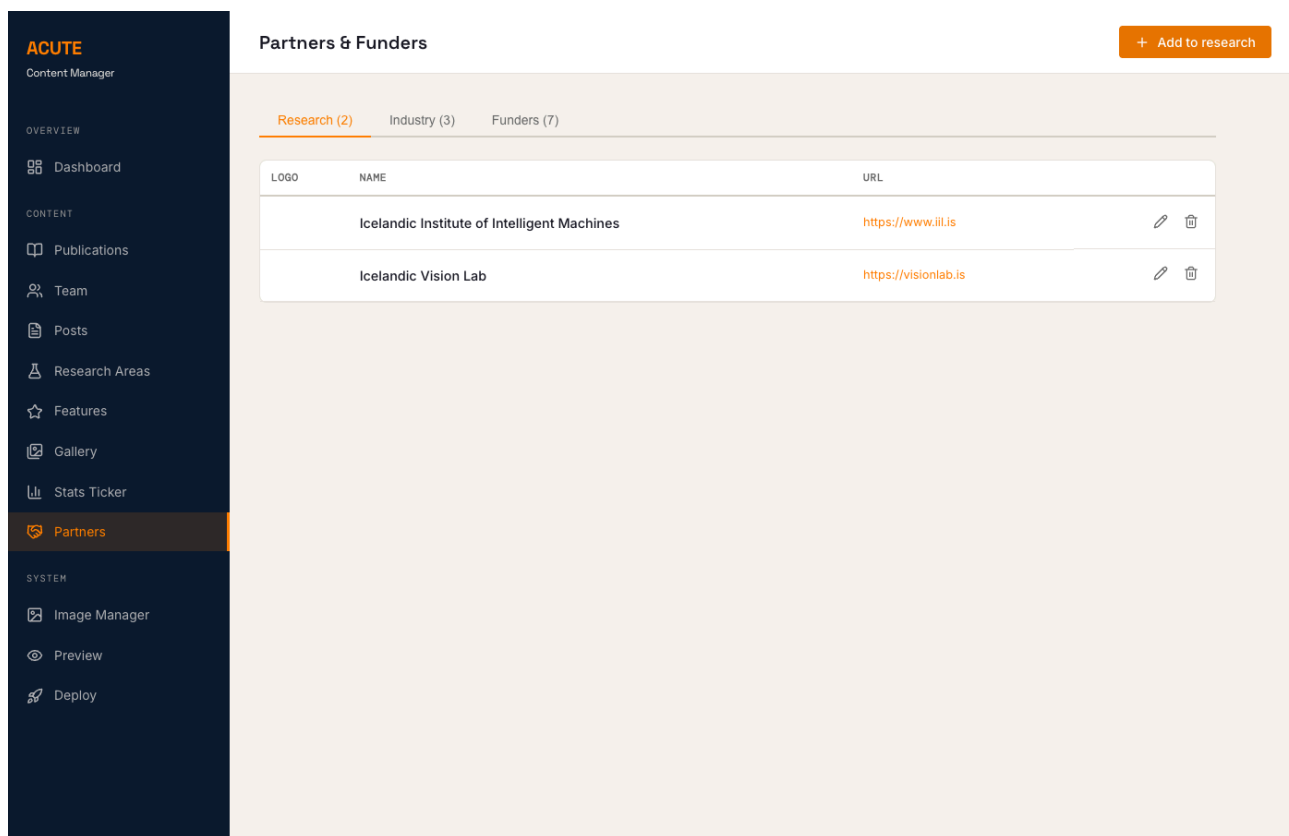
1. Click **+ Add Stat** (top right)
2. Enter the **Number** (e.g. "93%", "200 Hz", "34 dB")
3. Enter the **Label** (description of what the number means)
4. Click **Save**

Reordering

Use the up/down arrows to change the scroll order.

11. Managing Partners & Funders

Click **Partners** in the sidebar.



Partners are organized in three tabs:

- **Research** — academic research partners
- **Industry** — industry collaborators
- **Funders** — funding organizations

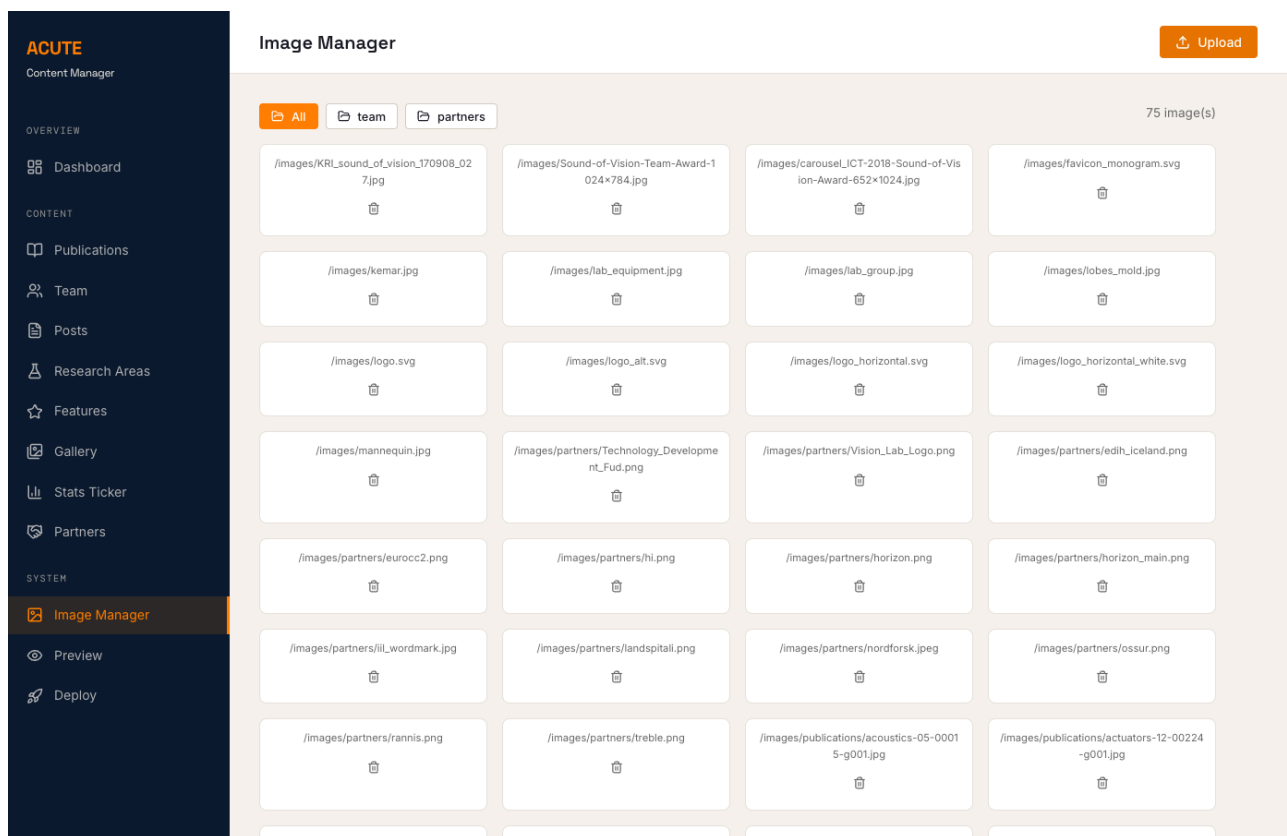
Adding a partner

1. Select the correct tab
2. Click the orange **+ Add to [group]** button
3. Fill in the name, website URL, logo (use Upload), and alt text
4. Click **Save**

Logos are auto-compressed to 300x150 pixels.

12. Image Manager

Click **Image Manager** in the sidebar.



Browsing images

Use the filter tabs at the top:

- **All** — every image on the site
- **team** — team member photos
- **partners** — partner and funder logos

Uploading images

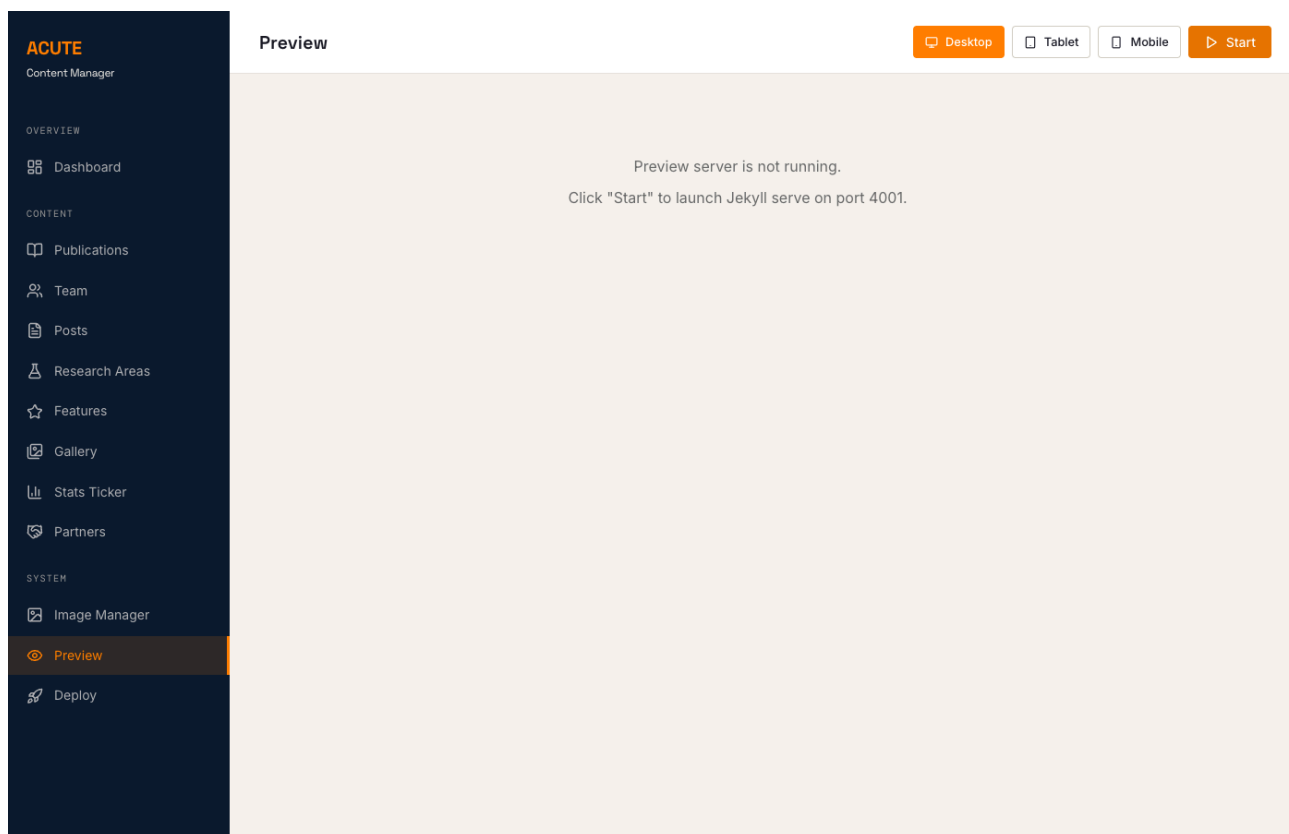
1. Click the orange **Upload** button (top right)
2. Select one or more files
3. Images are automatically compressed:
 - Team photos: max 400x400, JPEG
 - Partner logos: max 300x150, PNG
 - Other images: max 1200px wide, JPEG

Deleting images

Click the trash icon below any image and confirm.

13. Previewing the Website

Click **Preview** in the sidebar.



Note: This feature requires Ruby and Jekyll installed on your computer. This is optional — you can skip it and your changes will still work after publishing.

Installing Ruby + Jekyll (if you want Preview):

OS	How to install
Windows	Download RubyInstaller (Ruby+Devkit version). After install, open a new Git Bash and run: <code>gem install jekyll bundler</code> then <code>cd ~/Documents/acute_website_v2 && bundle install</code>
macOS	Ruby is pre-installed. Run: <code>gem install jekyll bundler</code> then <code>cd ~/Documents/acute_website_v2 && bundle install</code>

Using Preview:

1. Click the orange **Start** button
2. Wait for Jekyll to build the site (may take 10-20 seconds the first time)
3. The website appears in the preview area
4. Use the **Desktop / Tablet / Mobile** buttons to test different screen sizes
5. Click **Stop** when done

Every time you save content in the CMS, the preview updates automatically.

If you don't have Ruby installed, skip this step — just publish your changes and check the live website instead.

14. Publishing Your Changes

When you're happy with your edits, click **Deploy** in the sidebar.

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11 changed file(s)

Recent Commits

3904c9a

 - Responsive images: srcset with 480w/800w/1200w variants in images/responsive/ - Localized 5 publication thumbnails from MDPI CDN to images/publications/ - Inlined critical CSS (above-the-fold) and deferred main stylesheet loading - CSS minification step added to GitHub Actions deploy workflow - Added loading=lazy to research area images - Removed unused carousel_hrtf.heic (2.8 MB) - Stats ticker: expanded to 29 items, each linked to paper DOI - Research partners section added (ILL, Vision Lab) - Team page: collaborators group, past members (Hoffmann, Evarsson), Eric Sumner updated - Member layout: supports scholar and thesis links - Font size unification: consolidated into 3 tiers (0.8/0.85/0.9rem) - Mobile hero overflow fix - Blog/news system: _posts collection, post layout, /news/ page, homepage preview - Rúnar and Árni bios updated, PDF hosting in papers/ - Favicon updated to monogram SVG - Technology Development Fund added to funders - Removed CLAUDE.md from repo - Local CMS added (cms/): React + Express + TypeScript, manages all content types, image compression, Jekyll preview, one-click deploy

291c224

 Fixes and first deployment to labs github

638e4ca

 Go-live: compress images, add SEO/meta, data-driven refactor, new pages

Changed Files

Show Diff

? cms/docs/screenshots/01-dashboard.png

? cms/docs/screenshots/02-publications.png

? cms/docs/screenshots/03-publication-form.png

? cms/docs/screenshots/04-team.png

? cms/docs/screenshots/05-posts.png

? cms/docs/screenshots/06-gallery.png

? cms/docs/screenshots/07-stats.png

? cms/docs/screenshots/08-partners.png

? cms/docs/screenshots/09-research.png

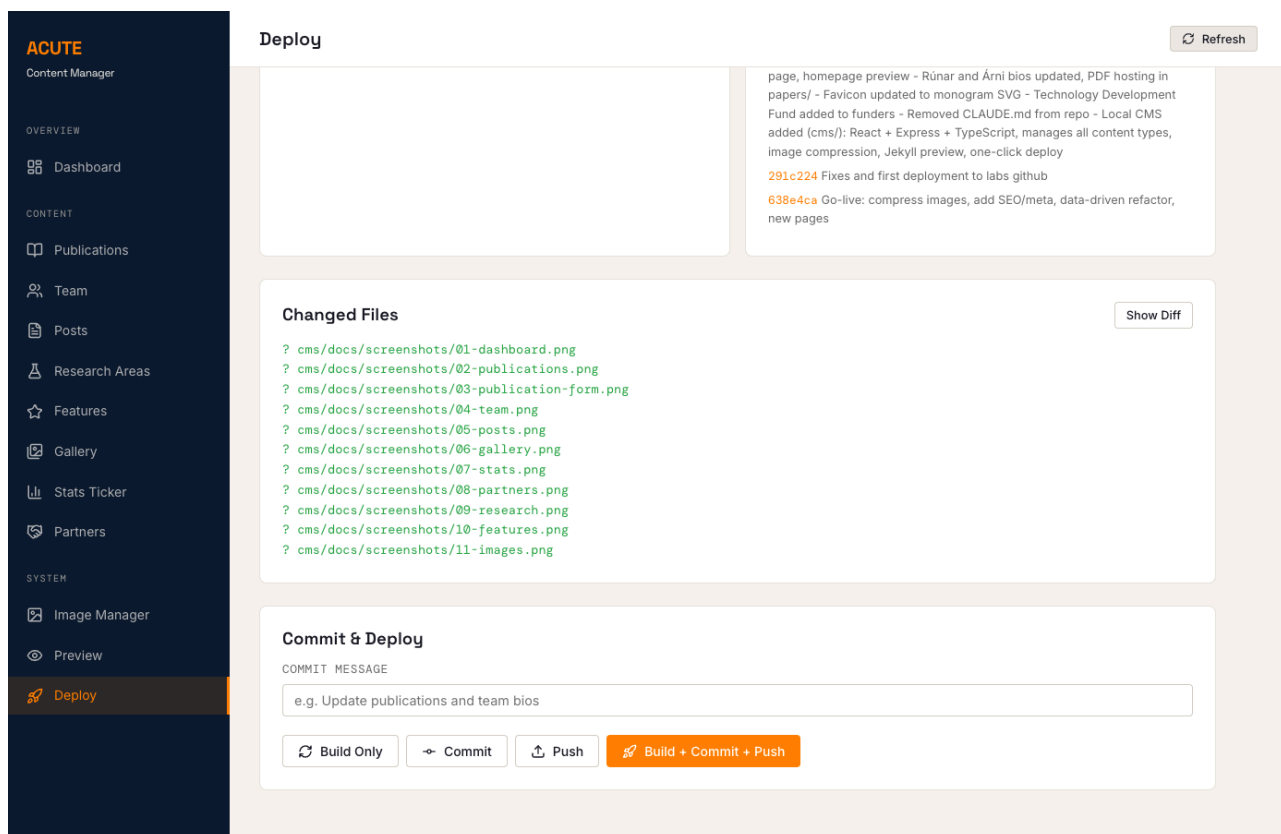
? cms/docs/screenshots/10-features.png

? cms/docs/screenshots/11-image.png

Step 1: Review your changes

- **Changed Files** shows everything you've modified
- Click **Show Diff** to see exactly what changed

Step 2: Commit and push



1. Type a short **commit message** describing your changes (e.g. "Add new publication by Karimi et al.")
2. Click **Commit** to save your changes locally
3. Click **Push** to publish to GitHub

Or use the orange **Build + Commit + Push** button to do everything in one click.

After pushing, the website updates automatically.

Important

- Always **pull the latest changes** before starting work (`git pull origin main` in terminal)
- Write clear commit messages so others know what changed
- Only one person should edit at a time to avoid conflicts

15. Troubleshooting

The CMS won't start

- Make sure you're in the `cms/` directory
- Run `npm install` again to ensure dependencies are up to date
- Check that Node.js 20+ is installed: `node --version`
- **Windows:** Make sure you're using **Git Bash**, not Command Prompt (cmd). The CMS may not work correctly in cmd.

Changes don't appear on the live site

- Make sure you committed AND pushed (check the Deploy page)
- Wait a minute — the site may take a moment to update after pushing

Preview doesn't work

- Preview requires Ruby and Jekyll installed locally
- If you don't have them, skip preview — your changes will still work after pushing

"Merge conflict" error when pushing

- This means someone else edited the same file at the same time
- Ask a team member with git experience to help resolve it
- To avoid this: always `git pull` before starting, and coordinate with colleagues

"Permission denied (publickey)" when pushing

- Your SSH key is not set up for GitHub
- Follow [Section 2: Setting Up SSH for GitHub](#) to create and add your key
- If you've already set up a key, try running `ssh-add ~/.ssh/id_ed25519` in your terminal, then try again
- If the issue persists, ask Stefanos for help

Images look wrong or don't upload

- Supported formats: JPEG, PNG, SVG, WebP
- Images are auto-compressed on upload — the original is not kept
- Maximum recommended size before upload: 10 MB

Quick Reference

Task	Where	Button
Add a publication	Publications page	+ Add
Add a team member	Team page	+ Add Member
Write a blog post	Posts page	+ New Post
Upload an image	Image Manager	Upload
Reorder items	Gallery / Stats / Features	Up/Down arrows
Preview the site	Preview page	Start
Publish changes	Deploy page	Build + Commit + Push

Website: <https://acute.hi.is>

Start the CMS: `cd cms && npm run dev` then open <http://localhost:3000>